

Appendix : Bayesian Inference and the Emergence of Internal Evaluation Structures : From Experience to Evaluation in Multi-Agent AI

Theoretical Premises : Systematic Correspondence with Bayesian Inference

A systematic consideration of the correspondence between the proposed experiment and Bayesian inference suggests that the proposed experiment can be interpreted not merely as measuring changes in an AI agent's internal representation, but as measuring the process by which the agent's internal state is updated in response to match experience treated as observational data.

Specifically, the transition

$$h^{pre} \rightarrow h^{post}$$

can be interpreted as analogous to the Bayesian transition

$$\text{prior} \rightarrow \text{posterior}$$

Under this interpretation, the latent representation h is treated as a representation of the agent's internal belief state.

This interpretation requires the following probabilistic structure.

Let D_{match} denote the experiential data acquired during a match.

Let $Q_{\text{pre}}(\theta)$ denote the agent's internal belief distribution concerning an underlying state or latent variable θ before the match, and let $Q_{\text{post}}(\theta)$ denote the corresponding belief distribution after the match.

$Q_{\text{pre}}(\theta) \rightarrow$ pre-match belief distribution

$Q_{\text{post}}(\theta) \rightarrow$ post-match belief distribution

We assume that the post-match belief is updated according to a Bayesian-style rule:

$$Q_{\text{post}}(\theta) \propto Q_{\text{pre}}(\theta) P(D_{\text{match}} \mid \theta)$$

Under this formulation, the pre- and post-match latent representations correspond to the respective belief states:

$$h^{pre} \leftrightarrow Q_{\text{pre}}(\theta) \quad \text{and} \quad h^{post} \leftrightarrow Q_{\text{post}}(\theta)$$

Consequently, the representational change

$$\Delta h = h^{post} - h^{pre}$$

can be interpreted not merely as a difference between two latent vectors, but as a latent-vector representation of the belief update induced by match experience.

This interpretation does not imply that the latent vector h is mathematically identical to the probability distribution $Q(\theta)$.

Rather, $\mathbf{h}$ is treated as a latent representation of the underlying belief state.

Under this interpretation, the proposed representational-change analysis provides an operational framework for investigating how experiential information acquired through interaction is incorporated into the agent's internal state. The magnitude, direction, and structural organization of $\Delta\mathbf{h}$ can therefore be examined as empirical signatures of experience-dependent belief updating.

No 1. $\mathbf{h}$ and the Belief State

Key point: $\mathbf{h}$ is treated as a latent representation of the agent's belief state.

The rationale for this interpretation is that, in conventional Bayesian inference, a belief state is generally represented as a probability distribution,

$$P(\theta \mid D_{\text{match}})$$

whereas the latent representation used in the present experiment is a vector,

$$\mathbf{h} \in \mathbb{R}^d$$

Therefore, $\mathbf{h}$ should not be identified directly with $P(\theta \mid D_{\text{match}})$, because the former is a vector representation whereas the latter is a probability distribution.

More precisely, $\mathbf{h}$ can be regarded as a latent representation of the underlying belief distribution, such that

$$\mathbf{h} = f(Q(\theta))$$

where $Q(\theta)$ denotes the agent's belief distribution and $f(\cdot)$ denotes a representation function that maps the belief distribution into the latent representation space.

Under this interpretation, $\mathbf{h}^{\text{pre}}$ and $\mathbf{h}^{\text{post}}$ represent the latent states corresponding to the agent's pre-match and post-match "beliefs", respectively.

No 2. $\Delta\mathbf{h}$ and Bayesian Updating

Key point: $\Delta\mathbf{h}$ provides a latent-vector representation of the Bayesian-style update.

The representational change is defined as

$$\Delta\mathbf{h} = \mathbf{h}^{\text{post}} - \mathbf{h}^{\text{pre}}$$

This quantity can be interpreted as an experience-dependent belief update represented in the latent space.

However, $\Delta\mathbf{h}$ should not be claimed to be mathematically identical to

$$P(\theta \mid D_{\text{match}}) - P(\theta)$$

The reason is that these two quantities belong to different mathematical spaces. $\Delta \mathbf{h}$ is a difference between latent vectors, whereas $P(\theta \mid D_{\text{match}}) - P(\theta)$ represents a difference between probability distributions.

Accordingly, the appropriate interpretation is that $\Delta \mathbf{h}$ provides a latent-vector representation of a Bayesian-style “belief” update, rather than constituting the Bayesian update itself in the strict probabilistic sense.

No 3. IRC and KL Divergence

Key point: IRC is analogous to a measure of Bayesian update magnitude.

The Internal Representation Change is defined as

$$\text{IRC}_i = \|\mathbf{h}_i^{\text{post}} - \mathbf{h}_i^{\text{pre}}\|_2 \quad \text{where } \|\cdot\|_2 \text{ denotes the Euclidean (L2) norm.}$$

Although IRC can be interpreted as a measure of the magnitude of experience-dependent representational change, it should not be regarded as mathematically equivalent to measures such as **KL divergence**, **Fisher information**, or **posterior contraction**.

For example, the **KL divergence** between the post-match and pre-match belief distributions is defined as

$$D_{\text{KL}} (Q_i^{\text{post}}(\theta) \parallel Q_i^{\text{pre}}(\theta))$$

where, D_{KL} denotes the Kullback–Leibler divergence.

which is mathematically distinct from

$$\|\mathbf{h}_i^{\text{post}} - \mathbf{h}_i^{\text{pre}}\|_2$$

Therefore, a stronger empirical correspondence with Bayesian inference may be obtained by measuring both quantities independently.

Specifically, in addition to the latent representational update,

$$\text{IRC}_i = \|\mathbf{h}_i^{\text{post}} - \mathbf{h}_i^{\text{pre}}\|_2$$

the experiment can introduce a Bayesian Update Magnitude (**BUI**),

$$\text{BUI}_i = D_{\text{KL}} (Q_i^{\text{post}}(\theta) \parallel Q_i^{\text{pre}}(\theta))$$

The relationship between **IRC** and **BUI** can then be examined empirically.

If a systematic positive relationship is observed, such that larger representational changes are associated with **larger Bayesian belief updates**,

$$\text{IRC} \uparrow \leftrightarrow \text{BUI} \uparrow$$

this would provide evidence that the observed changes in latent representation may reflect, or at least systematically correspond to, experience-dependent belief updating.

Such a result would strengthen the interpretation of the latent representational dynamics as an empirical signature of Bayesian-style updating.

No 4. SVD and the Bayesian Principal Update Direction

Key point: $\mathbf{V}_1$ should not be identified with a posterior principal direction. Rather, it represents the direction of maximum variance in the observed experience-dependent representational changes.

Given the collection of representational changes,

$$\Delta \mathbf{h}_1, \Delta \mathbf{h}_2, \dots, \Delta \mathbf{h}_N$$

SVD is applied to the corresponding data matrix,

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

The first right singular vector, $\mathbf{V}_1$ represents the direction in the latent representation space along which the observed representational changes exhibit the greatest variance.

Thus, $\mathbf{V}_1$ can be interpreted as the dominant direction of experience-dependent representational change: namely, the direction along which the AI agent's latent representations changed most substantially across match experiences.

Depending on the empirical relationship between these latent directions and the agent's observable behaviors, the resulting directions may be interpreted, for example, as

$\mathbf{V}_1$ = Cooperation Direction

$\mathbf{V}_2$ = Self-preservation Direction

$\mathbf{V}_3$ = Role-specialization Direction.

These semantic interpretations, however, are not determined by the SVD itself. SVD identifies statistically dominant directions in the latent representation space, but it does not intrinsically assign behavioral or semantic meanings to those directions.

The interpretation of a given direction therefore requires an additional analysis of its relationship with observable behavioral variables. For example, if the projection of an agent's representational change onto $\mathbf{V}_1$ is strongly correlated with measures of cooperative behavior, $\mathbf{V}_1$ may be interpreted as a Cooperation Direction.

Accordingly, the semantic meaning of the principal directions is not discovered by SVD alone; rather, it is identified empirically through the relationship between the latent directions and externally observable behavioral patterns.

This distinction is important because it separates the statistical extraction of latent directions from their behavioral interpretation, thereby providing a more rigorous basis for assigning strategic meaning to the resulting representational dimensions.

No 5. Cosine Similarity and Prior–Posterior Overlap

Key point: Cosine similarity measures the preservation of the direction of the latent representation, whereas KL divergence measures the difference between the underlying belief distributions.

The cosine similarity between the pre-match and post-match latent representations is defined as

$$\cos(h^{\text{pre}}, h^{\text{post}}) = (h^{\text{pre}} \cdot h^{\text{post}}) / (\|h^{\text{pre}}\|_2 \|h^{\text{post}}\|_2)$$

This measure quantifies the degree to which the directional structure of the latent representation is preserved following match experience. A high cosine similarity indicates that the representation remains oriented in a similar direction before and after the match, whereas a lower value indicates a greater change in representational orientation.

In contrast, the KL divergence,

$$D_{\text{KL}}(Q_i^{\text{post}} \parallel Q_i^{\text{pre}})$$

quantifies the difference between the post-match and pre-match belief distributions.

Thus, the two measures capture different aspects of experience-dependent change.

Cosine similarity addresses the question:

“To what extent is the direction of the latent representation preserved?”

In contrast, KL divergence addresses the question:

“To what extent has the underlying belief distribution changed?”

Accordingly, cosine similarity should not be interpreted as equivalent to prior–posterior overlap in the strict probabilistic sense. Rather, it provides a geometric measure of representational directional stability, whereas KL divergence provides a probabilistic measure of belief-state change.

The joint analysis of these two measures may therefore distinguish between changes in the geometry of the latent representation and changes in the underlying belief distribution. For example, a high cosine similarity combined with a large KL divergence would suggest that substantial belief updating has occurred while the overall direction of the latent representation has remained relatively stable.

No 6. ARI / NMI and Bayesian Posterior Clustering

Key point: ARI and NMI measure the stability of the organizational structure of the latent representation space before and after match experience.

The latent representations obtained before and after the match can be independently clustered, for example using **k-means**, yielding pre-match and post-match **cluster assignments**:

$$C_{pre}$$

and

$$C_{post}$$

The Adjusted Rand Index (**ARI**) can then be calculated as

$$ARI (C^{pre}, C^{post})$$

while the Normalized Mutual Information (**NMI**) can be calculated as

$$NMI (C^{pre}, C^{post})$$

These measures quantify the extent to which the organizational structure of the latent representation space is preserved across match experience.

In this context, **ARI** and **NMI** should be interpreted as measures of **representational structural stability**, rather than as direct measures of Bayesian posterior clustering. They do not themselves constitute Bayesian clustering procedures; instead, they provide empirical measures of whether the latent representational organization remains stable following experience.

A high ARI or NMI indicates that the cluster structure observed before the match is largely preserved after the match. Conversely, a low ARI or NMI indicates that the organizational structure of the latent representation space has undergone substantial reorganization.

In particular, a low value of **ARI (C^{pre}, C^{post})**, may indicate that match experience has not merely shifted individual latent representations, but has altered **the organization of the representational space itself**. Such a result would be consistent with the possibility that experience induces **a reorganization of the internal representational structure of the AI agents**.

Thus, whereas IRC measures the magnitude of individual representational change, and SVD identifies the dominant directions of representational change, **ARI** and **NMI** provide complementary measures of structural reorganization at the level of the representational space as a whole.

No 7. Integrated Bayesian Interpretation and the Emergence of an Inferred Evaluation Structure

Taken together, the systematic correspondence with Bayesian inference suggests that the proposed evaluation metrics should not be regarded as mathematically identical to their Bayesian counterparts. Rather, they can be understood as complementary measures that operationalize different aspects of Bayesian-style updating in the latent representation space.

In particular, the relationships among

IRC vs. KL divergence

SVD-derived v_1 vs. a principal direction of posterior updating
and

ARI/NMI vs. Bayesian clustering

represent conceptual correspondences rather than strict mathematical equivalences.

The addition of KL divergence provides an important probabilistic complement to the latent-vector measures.

The resulting framework can be organized as

IRC $\rightarrow$ KL divergence $\rightarrow$ SVD $\rightarrow$ Cosine similarity $\rightarrow$ ARI/NMI

corresponding respectively to

Magnitude $\rightarrow$ Bayesian Distance $\rightarrow$ Direction $\rightarrow$ Orientation $\rightarrow$ Structure

Under this framework, the proposed metrics characterize experience-dependent internal change at multiple levels: the magnitude of representational change, the magnitude of probabilistic belief updating, the dominant directions of change, the preservation of representational orientation, and the stability or reorganization of the latent representational structure.

7.1 Physical Experience and Social Evaluation

The proposed framework in the main text can be further extended to the role of the evaluation function **E**. In the present experimental setting, information such as MVP decisions and penalty decisions differs qualitatively from direct physical experience. Whereas physical interactions are generated during the match itself, MVP and penalty decisions are externally imposed evaluative signals received after the match. These signals can therefore be interpreted as forms of social or institutional evaluation. Accordingly, the match experience can be decomposed into two components:

$$D_{\text{match}} = \{ D_{\text{physical}}, D_{\text{social}} \}$$

where D_{match} denotes the total experiential data acquired in relation to a match.

Here, D_{physical} represents physical and interactional experience during the match, including contact, spatial position, cooperation with other AI agents, and attacks from opposing agents. In contrast, D_{social} represents socially or institutionally generated evaluative information, including MVP decisions, penalty decisions, and reward signals.

The post-match internal representation can therefore be expressed schematically as

$$h_{\text{post}} = F(h_{\text{pre}}, D_{\text{physical}}, D_{\text{social}})$$

where F denotes the agent's experience-dependent updating process.

This formulation highlights an important distinction from conventional reinforcement-learning settings. In a conventional reinforcement-learning formulation, an explicit reward or evaluation function is generally specified as part of the learning environment.

In the present framework, however, the evaluative structure may instead be treated as something that the agent must infer from interaction and externally generated social or institutional signals.

7.2 Inference of an Implicit Evaluation Function

Under this formulation, the agent can be considered to form an estimate of an evaluation structure E from its accumulated experience.

The corresponding inference can be represented as

$$P(E | D) \propto P(D | E)P(E)$$

where D denotes the accumulated experiential data and E denotes the underlying evaluation structure that the agent is attempting to infer.

The central question of the experiment can therefore be reformulated.

Rather than asking simply:

“ Did the AI learn? ”

the experiment asks:

“ Can an AI agent infer an evaluation structure that is not explicitly provided to it from socially mediated experience, and subsequently reorganize its internal representation in accordance with that inferred structure? ”

This formulation shifts the focus from learning as mere behavioral adaptation to the emergence of an internally represented evaluation structure through experience.

7.3 Sequential Formulation

When the experiment consists of repeated matches, the process can be represented sequentially as

$$D_{1:t} \rightarrow P(E \mid D_{1:t}) \rightarrow h_t \rightarrow \text{Behavior}_t$$

where D_t denotes the experiential data obtained at match t , $D_{1:t}$ denotes the accumulated experience from matches “1” through t , $P(E \mid D_{1:t})$ denotes the agent's inferred evaluation structure given its accumulated experience, h_t denotes its latent internal representation, and Behavior_t denotes its observable behavior at match t . The experience-dependent representational change can then be measured as

$$\Delta h_{t+1} = h_{t+1} - h_t$$

The magnitude of this change is quantified by

$$\text{IRC}_{t+1} = \|\Delta h_{t+1}\|_2$$

while the principal directions of representational change are obtained from the SVD of the collection of Δh vectors:

$$V_k = \text{principal directions of } \Delta h$$

The structural stability of the latent representation can be evaluated using

$$\text{ARI/NMI} = \text{Structural stability}$$

Finally, a central empirical relationship of interest is the correspondence between representational change and changes in the inferred evaluation structure:

$$\text{Bayesian update; } P(E \mid D_{1:t}) \text{ and, next match; } P(E \mid D_{1:t+1})$$

$$\Delta P_{t+1}(E) = P(E \mid D_{1:t+1}) - P(E \mid D_{1:t})$$

$$\Delta h_{t+1} \leftrightarrow \Delta P_{t+1}(E)$$

If a systematic relationship is observed between these quantities, it would provide evidence that changes in the AI agent's latent representation are associated with the progressive inference and revision of an evaluation structure from accumulated experience.

7.4 Notation

For clarity, the notation used throughout the framework is defined as follows.

θ = underlying state or latent variable about which the agent maintains a belief distribution

i = individual physical AI agent

t = match number / match round

N = total number of agent-match observations

N_{agents} = total number of physical AI agents

N_{matches} = total number of matches

d = dimensionality of the latent representation

When every agent participates in every match, the total number of agent-match observations is given by

$$N = N_{\text{agents}} \times N_{\text{matches}}$$

Thus, i identifies the individual physical AI agent, t identifies the match number, and N denotes the total number of agent-match observations used in the representational-change analysis.

As emphasized in the main text, the central proposal of this framework is not to provide agents with fixed prior preferences or a predefined evaluation function. Rather, the framework proposes that **the preference and evaluation structures themselves emerge through sustained interaction among agents within a multi-agent environment.** Thus, rather than assuming a fixed objective and asking how an agent can optimize it, the proposed setting asks how the objective itself can be inferred, negotiated, and progressively formed through social interaction.

I hope that this perspective may help conceptualize the framework as an extension from Bayesian belief updating under exogenously specified evaluative criteria to the endogenous formation of the evaluative structure itself through multi-agent interaction.

If individuality is defined as a distinctive pattern of behavior, meaning, and response that is formed through an AI agent's unique history of interactions, rather than as a fixed internal attribute, then **physical AI agents, like humans, may be capable of developing individuality through repeated interactions with their surrounding environments and other physical AI agents.**

Synthesis : At the end of this Appendix to the main paper, **Self-Awareness and Selfhood in Human and AI Cognition: Toward a Future Architecture for the Formation of High-Context Intelligence**, I would like to introduce a poem that can be regarded as a quintessential symbol of **"high context"**: a passage from the concluding section of Alfred, Lord Tennyson's *Ulysses*, which, in the film *Skyfall*, M, James Bond's superior, recites

"Though much is taken, much abides, and though we are not now that strength which in old days moved earth and heaven, that which we are, we are... One equal temper of heroic hearts, made weak by time and fate, but strong in will to strive, to seek, to find, and not to yield."

Ulysses (Odysseus) is a hero of Greek mythology who symbolizes wisdom, cunning, and perseverance. Tennyson constructed the poem as an embodiment of Victorian values: diligence, endurance, and active engagement. The poem's final four verbs, "strive / seek / find / not yield", constitute a concise expression of the Victorian ideal of "heroic perseverance."

Tennyson also establishes a clear distinction between Ulysses and his son Telemachus. Telemachus is recognized as suited to the role of a ruler whose task is to maintain order. Ulysses, by contrast, defines his own role not as governance but as "quest."

The passage may also be read as a final reflection on the framework presented here: intelligence and individuality are not necessarily fixed attributes, but may emerge through continued experience, interaction, and the pursuit of what lies beyond what is already known.

"Strive, seek, find, and not to yield" therefore serves as an appropriate closing thought—not as a conclusion to the inquiry, but as an invitation to continue it.

Godspeed on your research and development

With kindest regards,

Akira

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